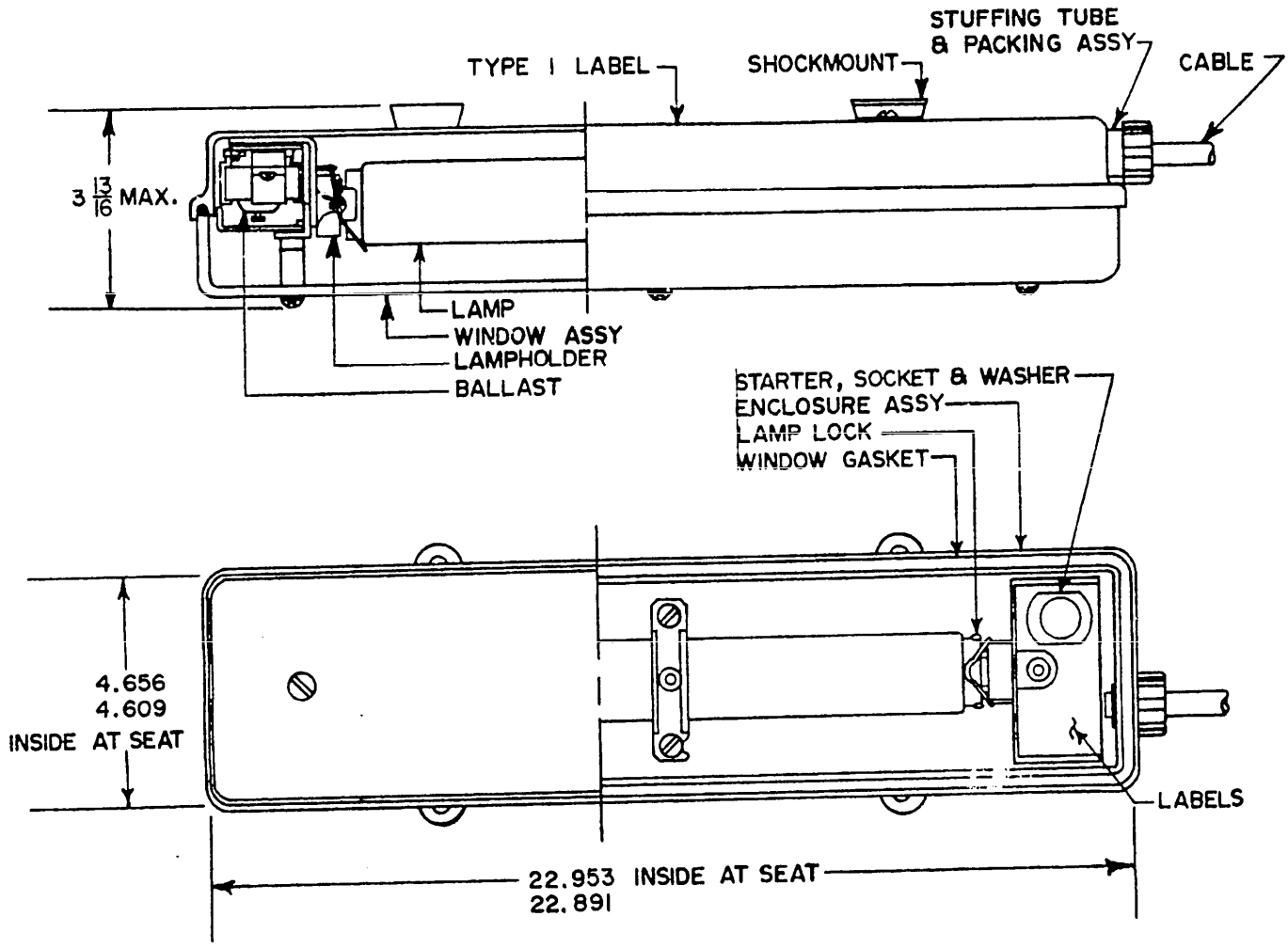
MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT, GENERAL LIGHTING,

15 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 338.1, 338.2, 339.1 AND 339.2

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SH).



SH8580 FIGURE 1. Typical view of Symbols 338.1 and 339.1.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

FSC 6210

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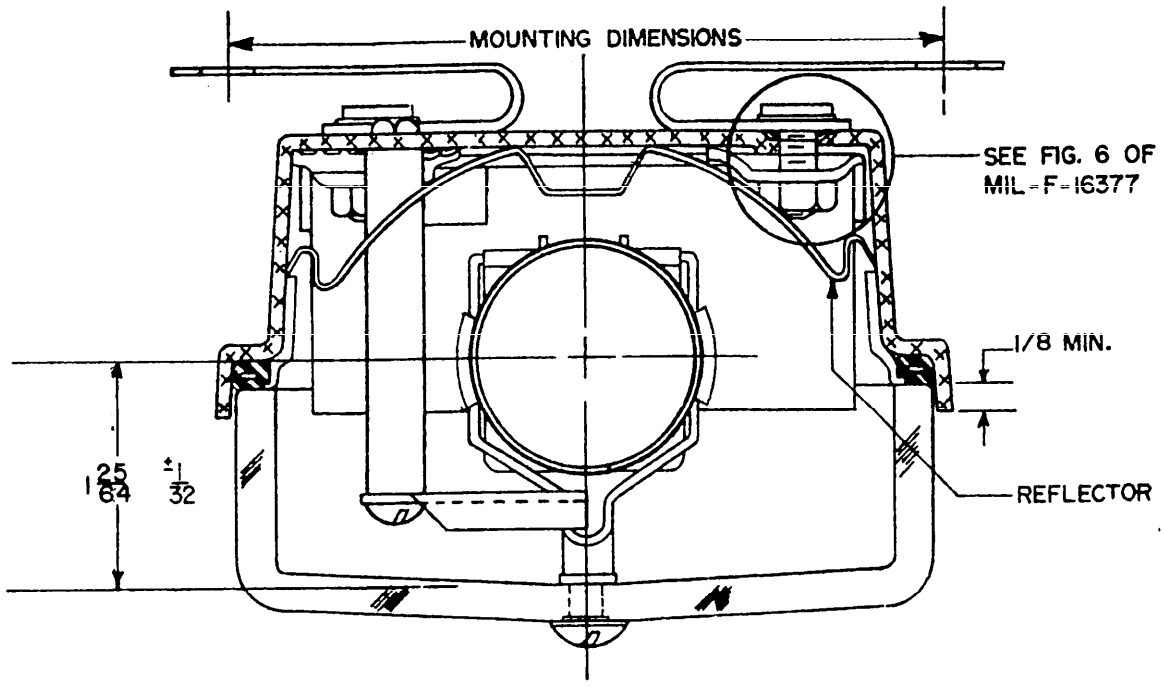


FIGURE 2. Typical section for Symbols 338.1 and 339.1.

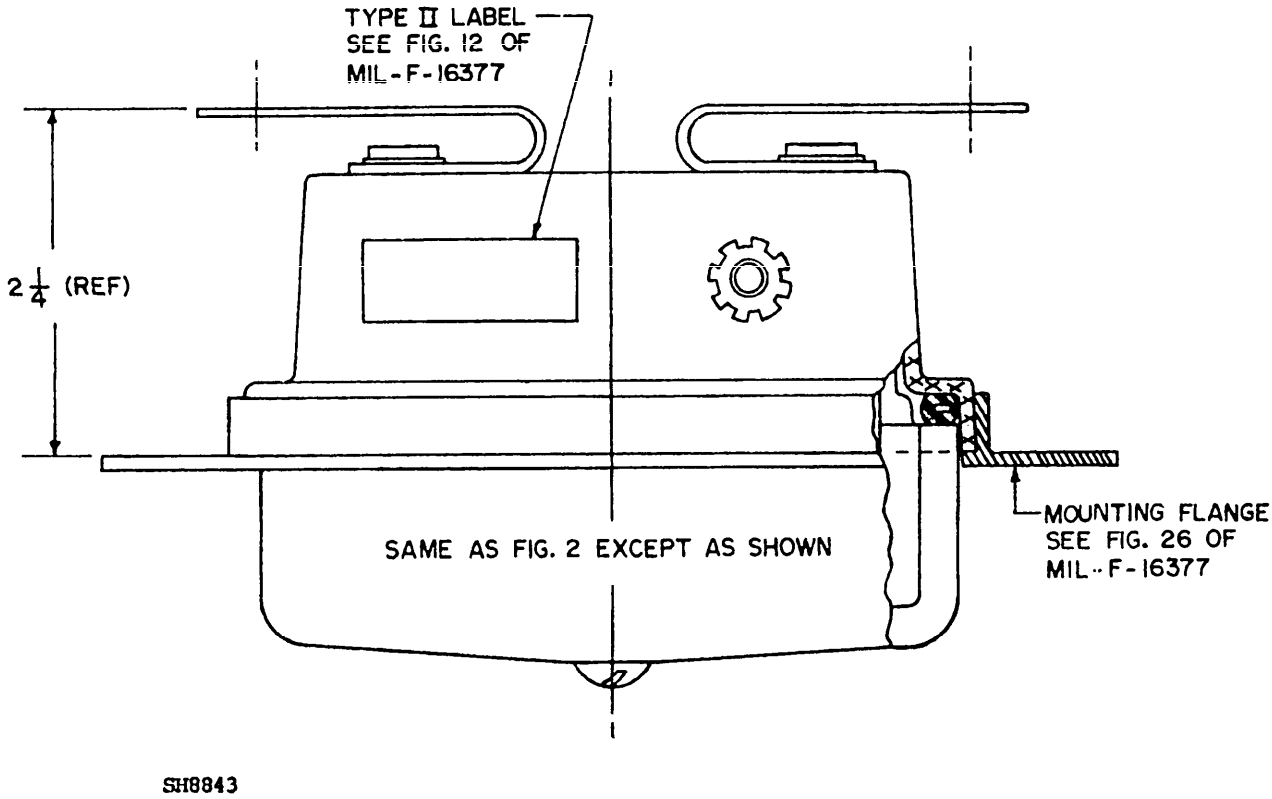
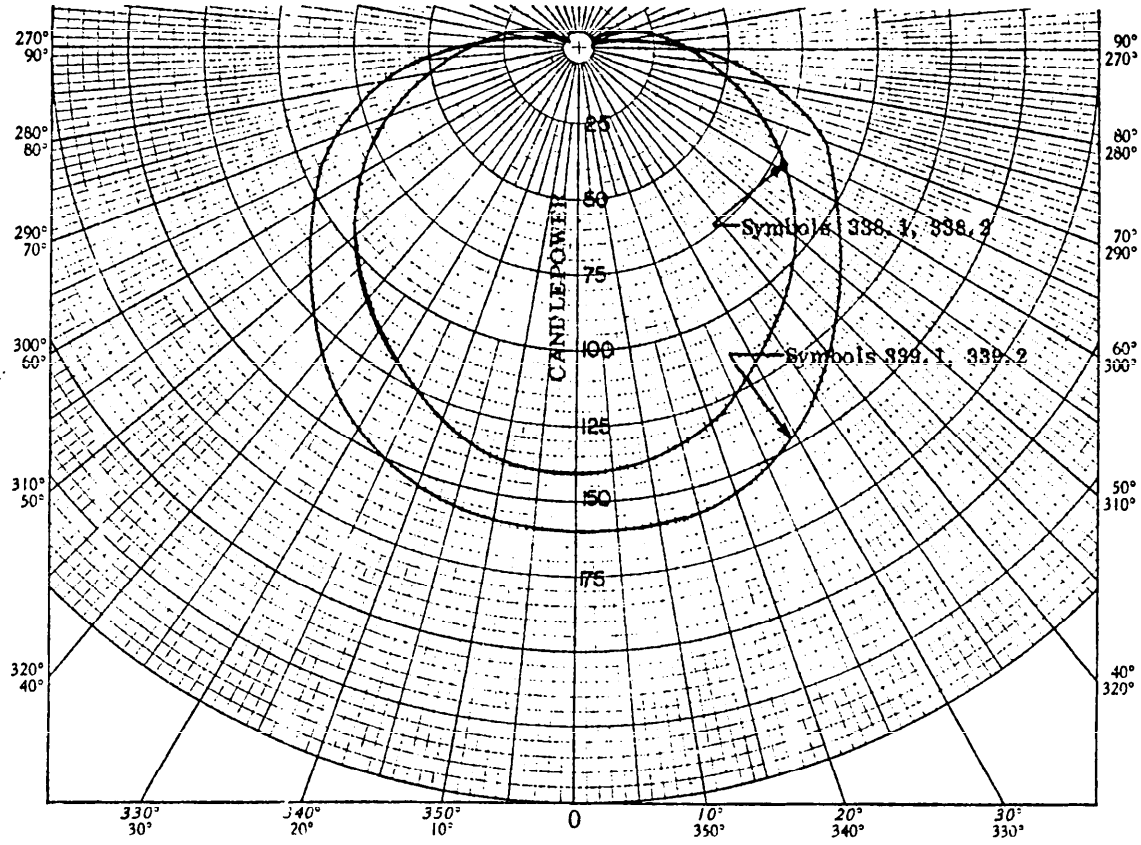


FIGURE 3. Typical section for Symbols 338.2 and 339.2.

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ANGLE	CANDLEPOWER SYMBOL NUMBER	
	338.1, 338.2	339.1, 339.2
0	137	158
5	137	157
10	135	156
15	133	154
20	130	151
25	125	147
30	120	145
35	115	141
40	108	135
45	103	126
50	96	114
55	90	104
60	83	98

ANGLE	CANDLEPOWER SYMBOL NUMBER	
	338.1, 338.2	339.1, 339.2
65	75	90
70	67	84
75	60	80
80	50	64
85	40	50
90	33	31
95	28	20
100	25	10
105	20	8
110	17	5
115	10	5
120	7	0
125	0	0

Photometric test data:

Distance: 10 feet.
Plane: At right angles to long axis of fixture through geometric center of lamp.
Voltage: 115 volts, 60 Hz.
Ambient temperature: 76°F.
Lamps: Use lamps furnished with fixtures seasoned for at least 100 hours.
Ballast: Use ballast furnished with fixture.
Lamp correction: Distribution curves based on 790 lumens per lamp.

FIGURE 4. Candlepower distribution curves.

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REQUIREMENTS:

CLASSIFICATION:
Type: I
Class: 2

SYMBOL NUMBERS:
338.1: Complete lighting fixture assembly, standard mounting, with one 15-watt cool white fluorescent lamp and white diffusing window assembly.
338.2: Complete lighting fixture assembly, flush mounting with one 15-watt cool white fluorescent lamp and white diffusing window assembly.
339.1: Complete lighting fixture assembly, standard mounting with one 15-watt cool white fluorescent lamp and clear and white prismatic window assembly.
339.2: Complete lighting fixture assembly, flush mounting, with one 15-watt cool white fluorescent lamp and clear and white prismatic window assembly.

DESIGN:
Electrical rating: 15 watts, 120 volts, 60 Hz.
Dimension and configurations: See figures 1, 2 and 3 and figure 10 of MIL-F-16377.
Photometric data: See figure 4.
Lampholder spacing: See figure 21 of MIL-F-16377.
Wiring diagram: See figure 22 of MIL-F-16377.
Mounting dimensions:
Symbols 338.1 and 339.1: See figure 25 of MIL-F-16377.
Symbols 338.2 and 339.2: See figure 26 of MIL-F-16377.
Window assembly:
Symbols 338.1 and 338.2: Military Part Number M16377/5-001, figure 27 of MIL-F-16377.
Symbols 339.1 and 339.2: Military Part Number M16377/5-002, figure 27 of MIL-F-16377.
Enclosure: Watertight. See 3.5.11 of MIL-F-16377.
Cable: 10 feet of type SJO or SJT, AWG size 16, two conductor (white and black) in accordance with J-C-580.
Weight:
Symbols 338.1 and 339.1: 6 lbs 12 oz. maximum.
Symbols 338.2 and 339.2: 8 lbs 3 oz. maximum.
Designation and marking: The following labels shall be furnished in accordance with MIL-F-16377 applied as shown in figures 1 and 3:

(a) Two type I.
(b) One type II for Symbols 338.2 and 339.2 only.
(c) One type IV. The information to be shown on the type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY." List all standard parts shown in table I and window assemblies of table IV by name and stock number only.

Standard Parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

QUALITY ASSURANCE: Quality assurance shall be as specified in MIL-F-16377 and as follows:

First article inspection:	
Complete fixtures:	See table II
Window assemblies:	See table III
Quality conformance:	
Comparison inspection:	Same as the first article inspection
Inspection of product for delivery:	
Examination:	See MIL-F-16377
Examination for preparation of delivery:	See MIL-F-16377
Quality conformance tests:	See MIL-F-16377

GENERAL INFORMATION:

Assignment of Military part numbers and general information is shown in table IV.

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TABLE I. Standard parts.

Name	Specification or Military part number	National Stock Number (NSN)	Remarks
Lamp	W-L-116/13	6240-00-583-3668	Industry Number F15T12/CW
Lamp lock	M16377/40-002	6250-00-399-7040	
Lampholder	MIL-L-970/11	6250-00-284-0481	
Startersocket	MIL-L-970/13	6250-00-295-2738	
Starter	8W, Type III of W-S-755	6250-00-299-2884	Commercial Designation: FS 2
Ballast	M16377/44-001	6250-00-569-9502	
Shockmount	M16377/41-009 of M16377/41-010	6210-00-973-1747 6210-00-973-1748	See 3.5.16 of MIL-F-16377
Red filter tube	M16377/42-004	6210-00-834-1621	Not furnished. See 3.7.7 of MIL-F-16377

TABLE II. First article and comparison inspection for complete fixtures.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Noise	3.5.14	4.8.9	
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Light output	3.7.1	4.8.6.1	Candlepower
Magnetic permeability	3.4.8	4.8.18	

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TABLE III. First article and comparison inspection for window assemblies.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Light output	3.7.1	4.8.6.1	Candlepower

TABLE IV. Part number and general information.

Military Part number M16377/5-	Item Name	National Stock Number (NSN)	Remarks
338.1	Complete fixture, Symbol 338.1	6210-00-725-3556	1/
338.2	Complete fixture, Symbol 338.2		
339.1	Complete fixture, Symbol 339.1		New
339.2	Complete fixtures, Symbol 339.2		New
001	Window assembly, white diffusing	6210-00-883-5592	2/
002	Window assembly, clear and white prismatic		New

1/ Previously covered by Drawing NAVSHIPS 815-1197146 and MS17193.

2/ Previously covered by MS16891, part No. MS16891-44.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6210-N553)

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5. PROBLEM AREAS

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b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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